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EDUCATIONAL QUALIFICATIONS:

Postdoctoral Scholar, University of Pennsylvania, Philadelphia, USA	04/2021-Present
Ph.D. in Biotechnology, National Centre for Cell Science, Pune, India	03/2021
M.Sc. in Biotechnology, Aligarh Muslim University, Aligarh, India	06/2013
B.Sc. in Zoology, Aligarh Muslim University, Aligarh, India	06/2010

PERSONAL STATEMENT:

The molecular mechanisms of cell proliferation and the field of the Ubiquitin Proteasome System (UPS) have always been my primary areas of interest and fascination. I possess the expertise, training, and motivation required to explore various facets of this incredibly vast field. Specifically, my training in mechanistic sciences, including molecular biology, biochemistry, immunology, virology, and cell/animal biology, has provided me with a strong foundation to comprehend how ubiquitin ligase complexes monitor the molecular mechanisms involved in cell cycle, viral infection and cancer. I've established a proximity labeling technique for identifying protein-drug interactions. My passion drives my continuous quest for new methods and scientific knowledge. My long-term career goal is to establish a research program as an investigator in cancer biology, dedicated to identifying novel cancer-specific proteins and utilizing targeted protein degradation for therapeutic applications. Additionally, I am equally enthusiastic about educating and mentoring the next generation of researchers.

RESEARCH EXPERIENCE:

Postdoctoral Fellow.: April 2021- Present

An important avenue of my research is the understanding of how E3 ubiquitin ligases contribute to disease development and progression to design effective personalized therapies. I am interested in studying the mechanisms by which the E3 ligases controls cell proliferation. In addition, I am exploring innovative approaches to chemically hijack E3 ligases for therapeutic purposes, such as through the development of Proteolysis Targeting Chimeras (PROTACs), which could lead to novel treatments that exploit the cell's natural degradation pathways to target and eliminate disease-driving proteins.

Ph.D.: March 2014- March 2021

During Ph.D., I worked on multiple projects related to role and regulation of E3 ubiquitin ligases in cell cycle progression, DNA damage response, HIV-1 infection, and cancer. I also screened and investigated anticancer properties of natural and synthetic chemical compounds.

The title of my thesis work is entitled as "Understanding the post-translational regulation of β -TrCP". It includes deciphering the mechanism governing the turnover of F-box protein β -TrCP1 by ubiquitination mediated proteasomal degradation during cell cycle. We demonstrated for the first time that β -TrCP1 is regulated through proteasome during G1/S phase of cell cycle by Cullin1-SKP1-FBXW8-Cullin7 E3 ligase complex. Through a series of mutational and biochemical studies, we delineated the synergistic association of Cullin1 and Cullin7 with FBXW8 and its importance

for the stability and ubiquitin ligase activity of the complex. Flow cytometric and immunofluorescence analysis showed that FBXW8 mediated β -TrCP1 regulation is important for S phase entry as well as error free S phase progression. We have also investigated the cross-regulation of β -TrCP1 and β -TrCP2. β -TrCP1 functions as tumor suppressor while β -TrCP2 acts as an oncogene. Tumor suppressor β -TrCP1 inhibits breast cancer tumorigenesis through attenuating oncogene β -TrCP2 expression levels. We also found that β -TrCP1 interfere the function of β -TrCP2 during DNA damage, leading to p53 activation and cell cycle arrest.

Postgraduate (M.Sc.) Research Training:

I have worked on the research project entitled “Protective effect of thymoquinone on glucose or methylglyoxal-induced glycation of superoxide dismutase”. This project included glycation of superoxide dismutase by glucose or methylglyoxal in absence and presence of thymoquinone. Next, we checked activity by enzyme assay and the amount of protein on SDS-PAGE gels as well as formation of high molecular weight aggregates. This study revealed that thymoquinone is an anti-glycating compound and can be effective against diabetes and its complications.

Selected Publications:

1. **Islam S**, Gour J, Beer T, Tang HY, Cassel J, Salvino JM, Busino L. A Tandem-Affinity Purification Method for Identification of Primary Intracellular Drug-Binding Proteins. *ACS Chem Biol*. 2024 Feb 16;19(2):233-242.
2. Sahay O, Barik GK, Dey S, **Islam S**, Paul D, Bhavsar PP, Nath S, Rapole S, Santra MK. Loss of FBXO31-mediated γ H2AX foci formation impairs initiation of NHEJ and HR repair pathways, and sensitizes breast cancer to therapy. *bioRxiv* 2024.06.03.596644; doi: <https://doi.org/10.1101/2024.06.03.596644>.
3. Barik GK, Sahay O, **Islam S**, Ghate NB, Kalita B, Alam A. Ubiquitination in Cancer Metastasis: Emerging Functions, Underlying Mechanisms, and Clinical Implications. *Technol Cancer Res Treat*. 2023 Jan-Dec;22:15330338231210720. (**Editorial**)
4. Rasool RU, O'Connor CM, Das CK, Alhusayan M, Verma BK, **Islam S**, Frohner IE, Deng Q, Mitchell-Velasquez E, Sangodkar J, Ahmed A, Linauer S, Mudrak I, Rainey J, Zawacki KP, Suhan TK, Callahan CG, Rebernick R, Natesan R, Siddiqui J, Sauter G, Thomas D, Wang S, Taylor DJ, Simon R, Cieslik M, Chinnaiyan AM, Busino L, Ogris E, Narla G, Asangani IA. Loss of LCMT1 and biased protein phosphatase 2A heterotrimerization drive prostate cancer progression and therapy resistance. *Nat Commun*. 2023 Aug 29;14(1):5253.
5. Barik GK, Sahay O, Mukhopadhyay A, Manne RK, **Islam S**, Roy A, Nath S, Santra MK. FBXW2 suppresses breast tumorigenesis by targeting AKT-Moesin-SKP2 axis. *Cell Death Dis*. 2023 Sep 22;14(9):623.
6. Sahay O*, Barik GK, **Islam S***. FBXW10: a male-biased E3 ligase in liver cancer. *Trends Cancer*. 2023 Nov;9(11):876-878 (***Correspondence**)
7. Agrawal Y, Sharma T, **Islam S**, Nadkarni KS, Santra MK. F-box protein FBXO41 suppresses breast cancer growth by inducing autophagic cell death through facilitating proteasomal degradation of oncogene SKP2. *Int J Biochem Cell Biol*. 2022 Jun;147:106228.
8. **Islam S**, Dutta P, Sahay O, Gopalakrishnan K, Muhury SR, Parameshwar P, Shetty P, Santra MK. Feedback-regulated transcriptional repression of FBXO31 by c-Myc triggers ovarian cancer tumorigenesis. *Int J Cancer*. 2022 May 1;150(9):1512-1524.
9. **Islam S**, Dutta P, Chopra K, Sahay O, Rapole S, Chauhan R, Santra MK. Co-operative binding of SKP1, Cullin1 and Cullin7 to FBXW8 results in Cullin1-SKP1-FBXW8-Cullin7 functional complex formation that monitors cellular function of β -TrCP1. *Int J Biol Macromol*. 2021 Nov 1;190:233-243.
10. **Islam S**, Dutta P, Chopra K, Rapole S, Chauhan R, Santra MK. FBXW8 regulates G1 and S phases of cell cycle progression by restricting β -TrCP1 function. *FEBS J*. 2021 Sep;288(18):5474-5497.
11. **Islam S**, Dutta P, Sahay O, Santra MK. β -TrCP1 facilitates cell cycle checkpoint activation, DNA repair, and cell survival through ablation of β -TrCP2 in response to genotoxic stress. *J Biol Chem*. 2021 Jan-Jun;296:100511.
12. Dutta P[#], **Islam S[#]**, Choppa S, Sengupta P, Kumar A, Kumar A, Wani MR, Chatterjee S, Santra MK. The tumor suppressor FBXO31 preserves genomic integrity by regulating DNA replication and segregation through precise control of cyclin A levels. *J Biol Chem*. 2019 Oct 11;294(41):14879-14895. (**#Authors contributed equally**)

13. Paul D, **Islam S**, Manne RK, Dinesh US, Malonia SK, Maity B, Boppana R, Rapole S, Shetty PK, Santra MK. F-box protein FBXO16 functions as a tumor suppressor by attenuating nuclear β -catenin function. J Pathol. 2019 Jul;248(3):266-279.
14. Augustine T, Chaudhary P, Gupta K, **Islam S**, Ghosh P, Santra MK, Mitra D. Cyclin F/FBXO1 Interacts with HIV-1 Viral Infectivity Factor (Vif) and Restricts Progeny Virion Infectivity by Ubiquitination and Proteasomal Degradation of Vif Protein through SCFcyclin F E3 Ligase Machinery. J Biol Chem. 2017 Mar 31;292(13):5349-5363.

EXPERIMENTAL AND TECHNICAL SKILLS:

Animal experiments: I have taken a basic course in animal handling during my Ph.D. course work studies and have experience in xenograft tumor formation in NOD-SCID mice.

Molecular biology: I have expertise in CRISPR, CHIP-Seq, RNA-Seq, Agarose gel electrophoresis, Polyacrylamide gel electrophoresis (Native and SDS-PAGE), western blotting, RNA isolation, Isolation of chromosomal and plasmid DNA, PCR, restriction digestion, ligation of the DNA and gene cloning by TA cloning and infusion cloning, generation of site directed mutants, cDNA, preparation, and real time PCR, Comet assay, DNA fragmentation assay. Expression and purification of recombinant proteins in bacteria.

Cell biology: Flow cytometry: BDCalibur and BDCanto II instrument for cell cycle analysis, BrDU pulse chase assay to monitor S phase, trained to use immunofluorescence and confocal microscopes independently, Immunostaining of fixed cells, live cell imaging, Immuno-precipitations studies, Nuclear-cytoplasmic and Chromatin fractionation, Senescence-associated beta-galactosidase staining.

Tissue Culture: I am experienced in handling and experimentation using cancer and primary cell line (suspension and adherent cultures). Freezing and reviving cells, maintenance of culture, generation of stable cell lines using viral infection, transfection, infection, synchronization of cells using different reagents including double thymidine, hydroxyurea, nocodazole, RO3306, aphidicolin and serum starvation, cell counting using hemocytometer. I can also perform Trypan blue exclusion assay, Colony formation assay, Soft agar assay, Cell migration assay, Bioluminescence assay, MTT assay.

Microbiology: Preparation of culture media & enumeration of bacterial colonies, Culturing of bacteria, preservation of cultures, bacterial transformation, ultra-competent cell preparation and Gram staining, Antibiotics susceptibility test by Standard Disc diffusion method.

Biophysics and Biochemistry: Qualitative and quantitative estimation of proteins and nucleic acids, enzyme activity assays Enzyme activity and Enzyme immobilization, Chromatography, Size fractionation by density gradient centrifugation, Circular dichroism, UV-VIS, Fluorescence Spectroscopy.

Immunology: Precipitation test, and agglutination test, Immunoelectrophoresis, radial immunodiffusion, double immunodiffusion, and ELISA.

Bioinformatics: R programming, BLAST, FASTA, Clastal W, UCSC genome Browser, Promoter sequence analysis, Ensembl database and Image J.

Statistical Tests: Various statistical tests for the significance of experiment. Knowledge of using Sigma plot, Graph Pad and MS Excel.

TRAINING:

- Completed **Leica SP5 II confocal microscope** training conducted by NCCS Bio-imaging facility during June-July 2019.
- Attended “**ZEISS** on Your Campus Light Microscopy Workshop-From Basics to Confocal Microscopy” at NCCS, Pune (2018).
- Attended BD **FACS Canto II** training held at NCCS, Pune (2017).

MENTORSHIP:

I have successfully mentored and guided several master students through the entire dissertation process. This involved formulating research hypotheses, designing experiments, providing comprehensive training, analysing data, and assisting in writing project reports. In addition, I have also provided initial training to multiple doctoral students, contributing to their academic development, and enhancing their research capabilities.

HONORS AND AWARDS:

2024	American Cancer Society (ACS) Postdoctoral fellowship
2024	Rising Star Award for Outstanding Research and Publication from The Normal and Malignant Hematopoiesis Research Affinity Group (RAG) of the CHOP Research Institute, Philadelphia, USA.
2018	Awarded International Travel Grant by CSIR to attend EACR conference in Barcelona, Spain.
2016-2019	Senior Research Fellowship -from Department of Biotechnology, Govt. of India, New Delhi, India.
2014-2016	Junior Research Fellowship -from Department of Biotechnology, Govt. of India, New Delhi, India.
2013	Graduate Aptitude Test in Engineering (GATE) with All India Rank 10 , Indian institute of technology (IIT), India
2013	CSIR-UGC National Eligibility Test (NET) , CSIR, Ministry of human resources, Govt. of India.
2011-2013	M.Sc. Biotechnology Studentship , Department of Biotechnology, Govt. of India, New Delhi, through an All-India Competitive Examination conducted by AMU, Aligarh, India.

EDITORIAL BOARD MEMBER AND EDITORIAL ACTIVITIES:

- Guest Editor in Frontiers in Cell and Developmental Biology.
- Guest Editor in Technology in Cancer Research & Treatment Journal.
- Review Editor in Frontiers in Cell and Developmental Biology.
- Editorial Review Board Member of Technology in Cancer Research & Treatment Journal.

REVIEWER OF INTERNATIONAL JOURNALS:

- AACR-Molecular Cancer Research.
- Cell Cycle Journal.
- Cancer Cell International Journal.
- BMC Molecular and Cell Biology Journal.
- European Journal of Integrative Medicine.
- International Journal of Biological Macromolecules.
- Biochemistry and Biophysics Reports.
- Biochemical and Biophysical Research Communications.
- World Journal of Surgical Oncology.
- Life Sciences.
- BBA - Molecular Cell Research.
- Molecular Cancer Therapeutics

POSTER AND PRESENTATIONS:

- Delivered an oral presentation in **Annual Scientific Retreat** organized by **Penn Center for Genome Integrity (PCGI)**, University of Pennsylvania, Philadelphia, USA (2024).
- Delivered an oral presentation in **ACC Heme Malignancies Program/CHOP Research Institute Normal and Malignant Hematopoiesis Research Affinity Group Joint Retreat**, Philadelphia, USA (2024).

- Delivered an oral presentation in **international conference on “Disease Biology: Diagnostics and Therapeutics** organized by Department of Biotechnology, S.P. Pune University, Pune, India (2020).
- Presented a poster in **international conference on “Tracking Cancer: Detection and Monitoring, from Diagnosis to Therapy** held at Barcelona, Spain (2019).
- Presented a poster in **international conference on molecular signaling** at NCCS Pune, India (2019).
- Presented a poster in **Indo-US conference on ‘Transcription, Chromatin Structure DNA Repair and Genomic Instability**, at IISC Bangalore, India (2018).
- Presented a poster in **2nd Mini-Symposium on Cell Biology** held at NCCS Pune, India (2016).
- Presented a poster in **International Conference on Cancer Research: New Horizons** held at NCCS Pune, India (2015).
- Presented a poster on the topic **Biodiversity and its importance** at Dept. of Zoology, AMU, Aligarh, India (2011).

CONFERENCES/SEMINARS/WORKSHOPS ATTENDED:

- Participated in **Wellcome Trust/DBT India Alliance Science Communication Workshop (SciComm)** held at Indian National Science Academy (INSA), New Delhi (2017).
- Participated in **International Symposium on Breast Cancer Research** at NCCS, Pune (2017).
- Participated in **Mini-Symposium Metabolomics and Volatomics towards disease markers** at NCCS, Pune (2016).
- Attended **International Symposium** held at Department of Zoology, AMU, Aligarh (2013).
- Attended **Annual Symposium** at School of Biotechnology, JNU, Delhi (2012).
- Participated in the **Silver Jubilee Symposium** at Department of Biochemistry, AMU, Aligarh (2012).

MEMBERSHIP OF PROFESSIONAL BODIES:

- Life Member of Indian Science Congress.
- 2019-2021: Member, European Association for Cancer Research

LANGUAGES KNOWN:

- English
- Hindi
- Urdu
- Arabic

Links:

<https://www.linkedin.com/in/sehbanul-islam-ph-d-b868114a>

<https://scholar.google.co.in/citations?user=4Mb7TnMAAAAJ&hl=en>